Case 07: Design and Development of a Patient-Specific Porous Orbital Floor and Infraorbital Rim Reconstruction Implant

1. Introduction

The orbital floor and infraorbital rim are critical anatomical structures that support the eyeball, maintain orbital volume, and preserve facial symmetry. Defects in this region may result from orbital blow-out fractures, zygomaticomaxillary complex (ZMC) fractures, facial trauma, tumor resection, or congenital deformities.

Patient-Specific Implants (PSIs) are designed from CT scan data to accurately reconstruct the defect by restoring the patient's original anatomy. Compared with conventional implants, PSIs provide improved anatomical fit, reduced surgical adjustment, and enhanced functional and aesthetic outcomes.

2. Clinical Indications

- Orbital blow-out fracture
- Zygomaticomaxillary Complex (ZMC) fracture
- Midfacial trauma
- Tumor resection defects
- Congenital craniofacial deformities
- Revision orbital reconstruction

3. Objective

- To design a patient-specific orbital floor and infraorbital rim reconstruction implant using Autodesk Meshmixer.
- To reconstruct the orbital defect using patient-specific anatomical geometry.
- To incorporate a **Voronoi perforation pattern** for weight reduction and improved tissue integration.
- To generate a printable STL model suitable for titanium additive manufacturing.

4. Software Used

Step	Software
CT Segmentation	D2P / 3D Slicer
Implant Design	Autodesk Meshmixer
Voronoi Pattern Generation	Autodesk Meshmixer
Surface Refinement	Autodesk Meshmixer
STL Export	Autodesk Meshmixer
Additive Manufacturing	Selective Laser Melting (SLM)

Autodesk Meshmixer was used for mirror reconstruction, implant contour generation, Voronoi perforation design, surface refinement, and STL export.

5. Methodology

CT Scan (DICOM)

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Segmentation

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3D Skull Reconstruction

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Orbital Defect Identification

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Mirror Reconstruction

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Patient-Specific Implant Design

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Thickness Generation

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Voronoi Pattern Generation

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Fixation Hole Design

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Surface Smoothing & Refinement

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STL Export

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SLM Manufacturing

6. Design Features

Feature	Function
Patient-Specific Geometry	Matches the patient's orbital anatomy
Mirror Reconstruction	Restores missing orbital anatomy
Orbital Floor Support	Maintains orbital volume
Infraorbital Rim Reconstruction	Restores facial contour
Voronoi Perforation Pattern	Reduces implant weight while preserving mechanical strength
Multiple Screw Tabs	Stable fixation to surrounding bone
Smooth Peripheral Edges	Reduces soft tissue irritation

7. Materials Used

Component	Material	Reason for Selection
Orbital Reconstruction Implant	Ti-6Al-4V ELI (ASTM F136)	High strength, corrosion resistance, excellent biocompatibility, suitable for SLM
Alternative Material	Medical Grade PEEK	Lightweight and radiolucent
Fixation Screws	Titanium Alloy	Excellent biocompatibility and corrosion resistance

8. Design Specifications

Parameter	Specification
Implant Type	Patient-Specific Orbital Floor and Infraorbital Rim Reconstruction Implant
Design Method	Mirror Reconstruction Technique
Design Software	Autodesk Meshmixer
Material	Ti-6Al-4V ELI (ASTM F136)

Manufacturing Method	Selective Laser Melting (SLM)
Implant Thickness	0.6–1.0 mm
Pattern Type	**Voronoi Perforation Pattern**
Perforation Size	Variable (Patient-Specific)
Fixation Method	Titanium CMF Screws
Screw Hole Diameter	Compatible with 1.5–2.0 mm CMF screws
File Format	STL

9. Importance of the Implant

Clinical Importance	Benefit
Restores Orbital Volume	Maintains normal eye position
Supports the Eyeball	Prevents enophthalmos
Restores Facial Symmetry	Improves aesthetic appearance
Patient-Specific Fit	Reduces intraoperative implant modification
Voronoi Architecture	Reduces implant weight while maintaining structural integrity
Stable Screw Fixation	Provides secure fixation to surrounding bone

10. Additive Manufacturing Details

Parameter	Specification
Material	Ti-6Al-4V ELI Powder
Manufacturing Technology	Selective Laser Melting (SLM)
Layer Thickness	30–60 µm (Typical SLM Process)
Post Processing	Support Removal, Surface Finishing, Cleaning
Sterilization	Medical Grade Sterilization

11. Result

A patient-specific porous orbital floor and infraorbital rim reconstruction implant was successfully designed using Autodesk Meshmixer. The implant accurately reproduced the patient's orbital anatomy and incorporated a **Voronoi perforation pattern** to reduce weight while maintaining structural integrity. Multiple fixation tabs were integrated to facilitate stable screw fixation. The final design was exported as an STL file suitable for fabrication using Selective Laser Melting (SLM).

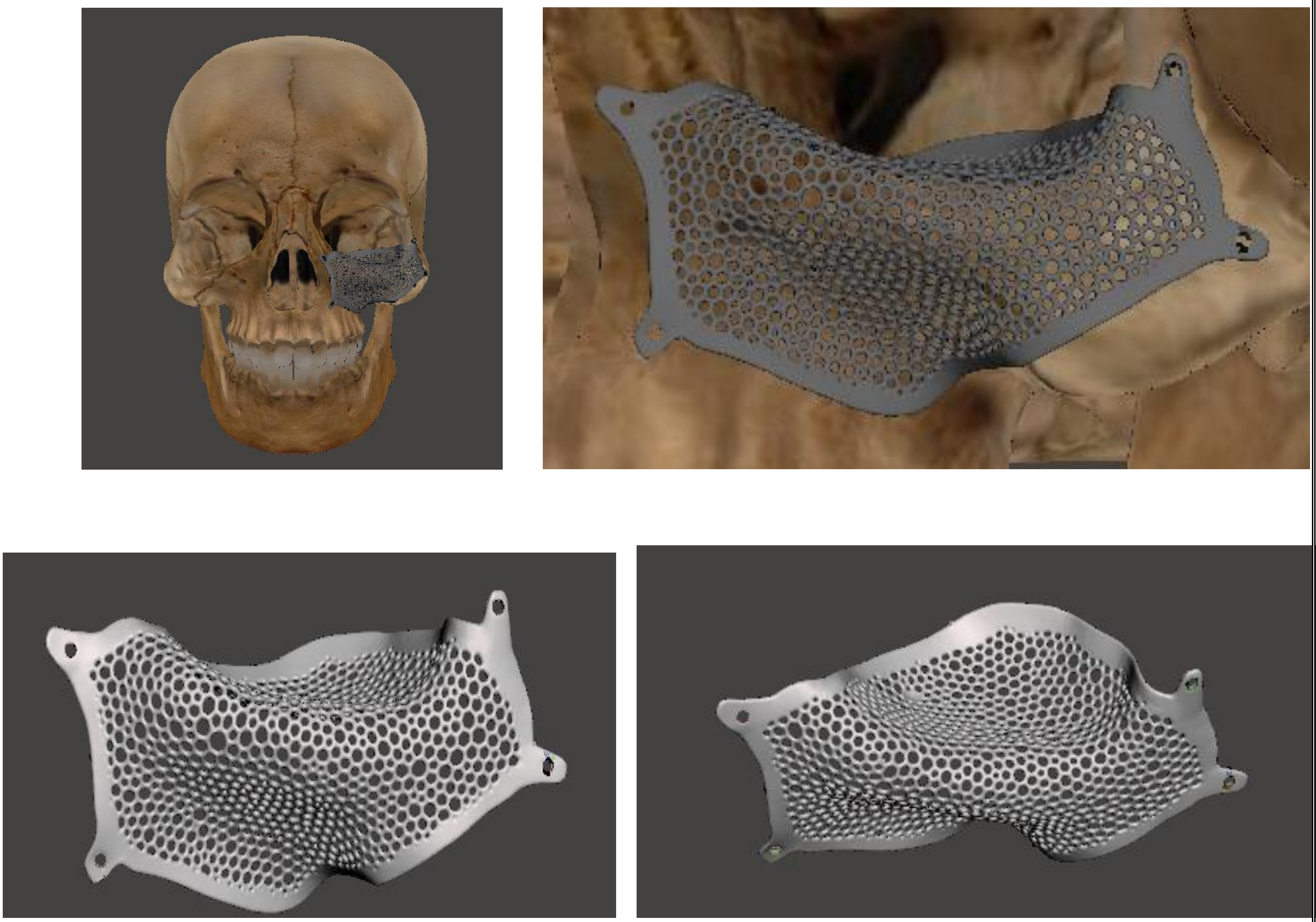

Figure. Patient-Specific Orbital Floor and Infraorbital Rim Reconstruction Implant with Voronoi Perforation Pattern designed using Autodesk Meshmixer and